

Minsuk Jang

Portfolio: tersite1.github.io
Github: github.com/tersite1

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Research Interests : Physical AI, Agentic AI, AI Safety

EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)** Daejeon, Korea
Master of Engineering - Electrical Engineering Expected Feb. 2027
- IPESK Next Generation Engineering researcher Award
- **Yonsei University** Seoul, Korea
Bachelor of Engineering - Civil and Environmental Engineering, AI Intensive; GPA: 3.78/4.30 Feb. 2025
Graduate Project: Estimation of Ansan's Forest Carbon Seizure Using SFM with LiDAR-Equipped Drones. (Best Project)
- Two Academic Excellence Scholarship

PROJECTS

- **DepthViz - Most Accurate Open Source SLAM Scanner on iPhone · Github (80 Stars) · Appstore**
: DepthViz is the first on-device iOS LiDAR SLAM app for infrastructure-free 3D mapping in Korea, solved blackbox problem on world's best-selling LiDAR. (*KR Patent 10-2024-0130327*), **Acquired to Nordic construction Startup**
- **Werkon - Smart Helmet System on Construction Site (Funded by KAIST KITS)**
: Werkon is a smart helmet system for construction managers equipped with 3D construction site reconstruction capabilities. It simultaneously detects and logs hazardous behavior. Presented at KAIST Innocore, Lotte hotel, Seoul

RESEARCH EXPERIENCE

- **Computational Intelligence Lab. - KAIST** Daejeon, Korea
Graduate Research Assistant (Advisor : Prof. Changick Kim) Feb. 2025 - Present
 - **Research Area:** 3D Reconstruction, Agentic AI, AI Robustness
 - **Thesis:** Real-Time 3D Digital Twin Reconstruction: A Lightweight LIO Framework Based on Mobile dTOF Sensors
- **Intelligence Unmanned System Lab. - Yonsei University** Songdo, Incheon
Undergraduate Research Assistant [Coursework] (Advisor : Prof. Jiwon Seo, Dr. Halim Lee) Jul - Sep 2024
 - **Research Area:** 3D Localization, Neural Radiance Fields, Autonomous Driving
 - **Publication:** Validating NeRF-Generated Point Clouds: A Comparative Study with KITTI Dataset for Accurate 3D Scene Reconstruction (**KONI 2024, Aug**)
- **Vehicle Intelligence Lab. - Seoul National University** Seoul, Korea
Undergraduate Research Assistant (Advisor : Prof. Seungwoo Seo) Apr - Jul 2024
 - **Research Area:** Autonomous Driving, Robotics Perception
 - **Supporting IROS Earth Rover Challenge 2024 [1st place]**
- **Autonomous Robot Intelligence Lab. - Seoul National University** Seoul, Korea
Undergraduate Research Assistant (Advisor : Prof. Seoungwoo Kim, Dr. Kyunghyun Kim) Jan - Mar 2024
 - **Research Area:** Robotics Perception, Neural Radiance Fields
 - **Publication:** VisionCycle: Automated Labeling Pipeline utilizing Neural Radiance Fields (**IEIE 2024, May**)

PROFESSIONAL EXPERIENCE

- **Idoll Robotics** Remote
Robotics Engineer - Building Ontology-based home agent robot Apr 2025 - Present
 - **Robotics Perception:** Implementation of user recognition and conversational capabilities for HomePod system
 - **Agentic AI:** Agent System Performing Multi-User Identification-Based Memory Control and Multi-Agent Orchestration
- **XYZ Innovation** Hybrid
Robotics Engineer Jul 2024 - Feb 2025
 - **E2E Drone System:** Development of Technology for Measuring Forest Carbon Sequestration Using Drones and LiDAR
 - **Two patents were sold to Stellarvision Inc.** (*Machine Learning Two patents were sold to Stellarvision Inc. Clustering-Based Method for Algal Bloom Detection in Satellite Imagery [10-2024-0165632] , Method for Generating a High-Precision 3D Model from a Digital Elevation Model [10-2024-0165631]*)
- **Stellarvision Inc.** Hybrid
AI Engineer - Drone Digital Twin Team Jul 2024 - Feb 2025
 - **Sensor Fusion:** 3D rendering of drone SAR images through LiDAR and SAR sensor fusion [**Exhibited at CES 2024**]
 - **Photogrammetry:** Digital Twin Construction of High-Risk Landslide Areas Using Drone Imagery
 - **Remotesensing:** Development of an AI model for oil spill detection using SAR and EO satellite imagery
- **Hyundai Motor Company** Hybrid
Undergraduate Researcher Scholarship Jan 2024 - Jul 2024
 - **LiDAR Percetpion:** Implementation of a LiDAR clustering algorithm for an international collegiate EV competition
 - **Control System:** Drone evasion maneuvering, landing algorithms for the ROK 2 Opt Command drone competition

PUBLICATIONS

Preprints / Under Review

- **MATE: Mobile Agent Trustworthy Execution for Safe and Efficient Mobile Control**
Minsuk Jang, Heeseon Kim, Huicheol Yu, Changick Kim
Recommendation at EMNLP Findings, **ACL Rolling Review May 2026** [Project Page]
- **Detection-Driven Image Super-Resolution for Mine-Like Object Detection**
Minsuk Jang, Byeongjun Park, Seho Kim, Kangook Ko, Changick Kim
Under review at **IEEE Transactions on Geoscience and Remote Sensing (TGRS)**
- **Spiking Neural Networks as Learnable Temporal Integrators for AI-Generated Video Detection**
Minsuk Jang, Yujin Yang, Hee-Seon Kim, Minseok Son, Younghun Kim, Changick Kim
arXiv:2605.05895 Under review at **NeurIPS 2026**
- **Gaussian Indexing: Multi-granular Instantiation for Open-Vocabulary 3D Scene Understanding**
Seho Kim, Inyong Koo, Minsuk Jang, Changick Kim
Under review at **NeurIPS 2026**
- **Masked AutoEncoders Are Latent Image Detectors**
Minsuk Jang, Hyunseo Jeong, Minseok Son, Changick Kim
arXiv:2511.06325 Under review at **NeurIPS 2026**

Published

- **LiTE ViT: Leaky Integrate-and-Fire Token Efficient Vision Transformer**
Minsuk Jang, Younghun Kim, Byeongjun Park, Changick Kim.
IEIE Summer Conference, 2025 (Oral) **BEST PAPER** (Top 3 out of 1400+ submissions)
- **Point Cloud Generation Using NeRF and Google Earth: Validation with the KITTI Dataset**
Minsuk Jang, Halim Lee, Jiwon Seo
Korean Navigation Institute (KONI) Fall Conference, 2024 (Oral)
- **Building Automation Labeling Pipeline Utilizing Neural Radiance Fields**
Minsuk Jang, Taehyung Kim, Shiho Kim
Institute of Electronics and Information Engineers (IEIE) Summer Conference, 2024 (Oral) [Project Page]
- **Quad-Tiltrotor VTOL Drone for Autonomous Search and Rescue: Design and Demonstration.**
Minsuk Jang Yoonchan Lee, June-Kyoo Park, Jihwan Shin, Sun-ho Song
Korea Robotics Society Annual Conference (KRoC), 2024, (link)

HONORS AND AWARDS

- **Excellence Award @ Next Generation Engineer Award, IPESK** Apr. 2026
Researcher award within Next Generation Engineers
- **Selected @ Next Generation Engineer, IPESK** Mar. 2026
Honor society for outstanding young engineers in Republic of Korea
- **1st Place @ 23rd AAM (Advanced Air Mobility) Tech Challenge, Korea AeroSpace Administration** Sep. 2025
Mission flight competition hosted by the Korea Aerospace Administration; development of LiDAR perception
- **Grand Prize (Governor's Award) @ World RoboFesta 2025, International Youth Robot Association** Jun. 2025
Drone-based End-to-End Forest Carbon Sequestration Estimation System
- **1st Place (Seoul Mayor's Prize) @ National Drone Utilization Competition, Seoul City** Dec. 2024
Drone-based Forest Digital Twin Accessible to Citizens
- **Grand Prize (Governor's Award) @ Jeju Satellite Data Challenge, Jeju Province** Sep. 2024
Algal bloom detection technology using remote sensing EO imagery

PATENTS

- LiDAR Micro Mobile Mapping System.
Minsuk Jang. [KR 1020260044350]
- Forest Carbon Sequestration Analysis System and Method with Drones.
Minsuk Jang. [KR 1020260066222]
- Vertically Takeoff and Landing Unmanned Aerial Vehicle with Adjustable Center of Gravity.
Minjun Kim, Junkyoo Park, Jungha Wang, Minsuk Jang. [KR 1020260086371]
- Method for Generating a High-Precision 3D Model from a Digital Elevation Model.
Minsuk Jang, Seungchul Lee. [KR 10-2024-0165631] **[Transferred to Stellarvision Inc.]**
- Machine Learning Clustering-Based Method for Algal Bloom Detection in Satellite Imagery.
Minsuk Jang, Seungchul Lee. [KR 10-2024-0165632] **[Transferred to Stellarvision Inc.]**

FUNDING & SCHOLARSHIPS

- KAIST GESS Funding Aug. 2026
- Brain Korea 21 Scholarship, Korea Government May 2026 - Present
- KAIST Startup & Innovation Center Scholarship Aug. 2025
- Hyundai Motor Company Undergraduate Research Scholarship Jan. - Jul. 2024
- Academic Excellence Scholarship, Yonsei University 2023 Fall, 2024 Spring